higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER PROGRAMMING

(First Paper)

NQF LEVEL 4

(10041024)

15 March 2019 (Y-Paper)
13:00–15:00

This question paper consists of 10 pages.

NCV

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TIME: 2 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 A special method that has the name New rather than Initialise

A Property
B Constructor
C Overload
D Function
E Method

1.1.2 … are used to model specialised attributes and behaviour.

A Subclasses
B Base classes
C Parameterised methods
D Parameterised functions
E Superclasses

1.1.3 … allow(s) one to refer to objects according to their superclass rather than their actual class.

A Encapsulation
B Inheritance
C Polymorphism
D Properties
E Functions

1.1.4 Database … is a process of changing the structure of a database to minimise the duplication of fields.

A record
B scope
C normalisation
D buffer
E tabling

1.1.5 To connect a database, Visual Basic must know the type of database which is stored in the … field.

A provider
B column
C name
D data source
E communication

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1.1.6 … enhance(s) the programmer's ability to understand program code.

A Unit testing
B Documentation
C Design
D Comments
E Error numbers

1.1.7 Money received from someone in return for using software you created:

A Freeware fees
B Copyright fees
C Royalty fees
D Service fees
E Design fees

1.1.8 Which ONE of the following is a version-control command to check files into the repository:

A Add-ins
B Admin
C Update
D Commit
E Transfer

1.1.9 Which ONE of the following is NOT the correct definition of an applet:

A Java program launched from HTML and runs in a browser
B Program designed to be executed from command prompt
C Small internet-based program written in Java
D Small software program that supports a larger application
E PHP Script to support data validation on a web site's "contact-us" form

1.1.10 When you pay a company to put your files on their web server, you are paying them for providing you a … service

A webhosting
B surveillance
C ownership
D royalty
E protection

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1.1.11 Which ONE of the following HTML tags does NOT belong to the header section of a webpage structure:

A <title>
B <meta>
C <link>
D
E <header>

1.1.12 The HTML attribute that causes the browser to display text if an image cannot be displayed.

A align
B href
C anchor
D alt
E substitute

1.1.13 "Error 404" is a standard HTTP status code that occurs when the Web Server …

A Does not have access to your local drive
B Is running under severe stress
C cannot find the specified page or coded resource link
D is running without a decryption certificate
E is running on an open source hosting platform

1.1.14 Which ONE of the following file extensions is an example of an animation file

A .mp3
B .wav
C .swf
D .png
E .jpg

1.1.15 … is a general-purpose Web-server application that runs on the Microsoft™ Windows platforms to serve requested HTML pages.

A Front Page™
B IIS™
C SQL Server™
D Dream Weaver™
E Go-Live™

(15 × 1) (15)

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1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 Programmers make instance variables private to enforce encapsulation.

1.2.2 A primary key is a relational database concept which states that table relationships must always be consistent.

1.2.3 Secure sockets layer (SSL) is the standard security technology for establishing an encrypted link between a web server and a browser.

1.2.4 The Uniform Resource Locator or URL, usually refers to the location of the user's desktop computer.

1.2.5 JavaScript has both strict and type-converting comparison operators.

(5 × 1) (5)

1.3 Give ONE term for each of the following descriptions. Write only the term next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Web content that contains a combination of different data or content formats such as audio, images, videos and animations

1.3.2 Standard mark-up language used to create documents on the World Wide Web

1.3.3 A data communication connection that uses a protocol in which no information is retained by either the sender or the receiver.

1.3.4 Microsoft's™ technology that assists with the sharing of application data and objects written in different formats from multiple sources.

1.3.5 Method of programming that focuses on objects rather than procedures

(5 × 1) (5)

[25]

TOTAL SECTION A: 25

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SECTION B

QUESTION 2

2.1 Consider the book and magazine objects presented in the class diagrams below:

Book
Title *Author* *Price* *Copies*
SellCopy() *OrderCopies()*

Magazine
Title *Price* *OrderQty* *CurrentIssue* *Copies*
SellCopy() *AdjustQty()*

Separate the common members of the classes into a superclass and the different members into subclasses. Use arrows to indicate the inheritance. (10)

2.2 List at least THREE physical resource specifications a web server must meet in order to support high traffic websites. (3)

2.3 Explain the difference between a website *storyboard* and a *wireframe*. (2 + 2) (4)

2.4 Your target audience has a lower-bandwidth internet connection.
Name TWO ways in which you could make your application use less bandwidth. (2)
[19]

QUESTION 3

3.1 The medical centre in your neighbourhood looks after the psychological, dental and general wellbeing of the community. You are part of the development team asked to design a website for them.

Use a flow diagram to draw a website sitemap in order to illustrate the general structure and hierarchy that the site will follow. (5)

3.2 Give THREE disadvantages of WYSIWYG text editors. (3)

3.3 Study the database table below and answer the questions that follow.

EMPLOYEE

EmpID	LastName	CellPhone
100	De Berg	064 455 9900
102	Naidoo	072 345 7706
103	Molepo	083 303 4012

3.3.1 Write a SQL statement to add a row into the table. The values are 104, Zulu, 072 664 0025. (3)

3.3.2 Write a SQL statement to change the values in the row with EmpID 100. The new values are Tempest and 082 100 3999. (3)

3.3.3 Write a SQL statement to delete the row with EmpID 102. (2)

3.4 Study the HTML file below and answer the questions.

```
Line 1 <html>
Line 2     <head><title>AGE CALCULATOR</title></head>
Line 3     <body onload = 'EnterYear()'>
Line 4          <script type = 'text/javascript'>
Line 5               function EnterYear() {
Line 6                    var age;
Line 7                    var birthYear;
Line 8                    birthYear=prompt('Enter your year of birth: ','YYYY');
Line 9                   age = (CurrentYear - birthYear);
Line 10                  alert(age);
Line 11              }
Line 12          </script>
Line 13    </body>
Line 14 <html>
```

3.4.1 When will the function in Line 5 be invoked? (1)

3.4.2 What will the code in Line 10 produce? (1)

3.4.3 Identify and correct the coding error in Line 9. (2)
[20]

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QUESTION 4

4.1 Compare and contrast *cohesion* with *coupling* in an OOP context. (3 + 3) (6)

4.2 Name THREE Internet technologies commonly used to create web applications. (3)

4.3 Consider the following HTML code and answer QUESTION 4.3.1 to 4.3.4.

```
No#     Code
1       <html>
2       <head>
3               <script scriptname='script.js'></script>
        </head>
4       <p>ABSA Premier League<p>
5       <table>
6               <tr><td>Played</td><td>Won</td><td><td>Points</td></tr>
7       </table>
8       <p>View the top goal scorers <a href='goals.html'> Click here
        </a><p>
        </html>
```

4.3.1 Correct the code in line 3 so that the external JavaScript code will be properly embedded in the listed HTML file. (2)

4.3.2 Change the code in line 4 so that 'ABSA Premier League' is a heading at level 2. (1)

4.3.3 Give the HTML tag that will turn the column content in Line 6 into column headers instead. (1)

4.3.4 You are part of a development team and responsible for creating goals2017.html.

(a) Describe what will happen when the web browser executes line 8.

(b) Change line 8 so that the browser does NOT open the file in a new window.

(2 × 1) (2)

4.4 What is the difference between *ownership* and *copyright* with regard to the software installed on a user's PC. (2 + 2) (4)

[19]

QUESTION 5

5.1 Define each of the following concepts:

5.1.1 Data binding

5.1.2 Dataset

5.1.3 Data adaptor (3 × 1) (3)

5.2 To access a webpage on a web server, you must enter the URL of the web site the address bar of the web browser. An example of a URL is http://tvet.edu.ac.za/notes/cssnotes.html.

Explain the different parts of this URL. (4)

5.3 The Life Orientation lecturer at your college is keeping track of her students' assessment marks by using Microsoft Excel.

Sample data

Student number	Student name	Phone	Task 1	Task 1 Mark	Task 2 Type	Task 2 Mark
201502	F Nare	0784563434	Assignment	87	Test	45
201623	H Tutu	0832345678	Assignment	66	Test	59
201456	K Smit	0612345678	Assignment	46	Test	19

Split the given spreadsheet into separate database tables. Indicate the names of fields, foreign and primary keys for each table. (3 × 2) (6)

5.4 Briefly explain each of the following forms of access control and authentication on the server side:

5.4.1 Challenge-response

5.4.2 One-time password (OTP)

(2 × 2) (4)
[17]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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